Sai Vikram Karna Ananth Balasingam

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Professional Summary

MS Data Science student at the University of Maryland, College Park with hands-on experience across machine learning, deep learning, data engineering, NLP, statistical analysis, and software development. I've built applied projects using Python, SQL, PyTorch, Spark, Kafka, Dask, Neo4j, Chroma, and cloud-oriented workflows — the kind of work where you learn as much from debugging a silent schema mismatch as you do from a model that trains cleanly on the first try. I'm looking for Data Scientist, Data Analyst, Machine Learning Engineer, Software Engineer, and Data Engineer roles where I can build reliable data products, analytical systems, and ML-driven solutions that actually get used.

Education

University of Maryland, College Park College Park, MD
Master of Science in Data Science *Aug 2024 – May 2026*
Easwari Engineering College, affiliated with Anna University Chennai, India
Bachelor of Engineering in Computer Science *Nov 2020 – 2024*

- Served as Secretary of the Computer Science Department and was an active member of the Association of Computer Engineers from 2021 to 2022 — organized events, coordinated with faculty, and learned that herding 200 students toward a deadline is excellent project management practice.

Technical Skills

Programming: Python, SQL, R, Java, C++, JavaScript, HTML, CSS
Machine Learning & AI: Scikit-learn, PyTorch, TensorFlow/Keras, Deep Learning, NLP, Generative AI, Time Series, Model Evaluation
Data Engineering: Apache Spark, PySpark, Kafka, Spark Structured Streaming, Dask, Dask-ML, ETL, Data Pipelines
Data & Analytics: Pandas, NumPy, Matplotlib, Statistical Analysis, Hypothesis Testing, EDA, Tableau, Power BI
Databases & Tools: MySQL, MongoDB, Neo4j, Chroma Vector Database, Git, GitHub, Docker, Jupyter, Google Colab

Experience

PluginLive Chennai, India
Data Science Intern *Jul 2023 – Dec 2023*

- Built and cleaned structured datasets using Python, Pandas, NumPy, and SQL to feed into analytics, reporting, and machine learning workflows — spent a lot of time hunting down why certain fields were null in one source and populated in another, which taught me that data cleaning is rarely a one-pass job and almost always reveals something surprising about how the data was collected in the first place.
- Ran exploratory data analysis to surface trends, flag data-quality issues, and pull out business insights that informed product and operational decisions — the kind of work where you start with a vague question (”are users in this segment behaving differently?”) and end with a concrete answer and a chart that makes it obvious.
- Designed and ran machine learning experiments using Scikit-learn — handled preprocessing, feature engineering, model training, evaluation, and performance comparison across multiple approaches. Learned that model selection matters less than feature quality, and that a simple model with clean features usually beats a complex model fed messy data.
- Wrote reusable analysis scripts and documentation so technical and non-technical stakeholders could understand the outputs without me standing over their shoulder explaining every chart — the documentation paid for itself the first time someone used a notebook I'd written three months earlier and didn't need to ping me with questions.

Skolar Chennai, India
Data Science Intern *May 2022 – Oct 2022*

- Worked on the full data science workflow — data preprocessing, exploratory analysis, model building, and visualization — using Python, Pandas, NumPy, Matplotlib, and Scikit-learn. This was where I built the muscle memory for the routine steps (checking for nulls, plotting distributions, scaling features) so I could focus brainpower on the harder parts like feature engineering and interpreting results.
- Applied supervised machine learning techniques to analyze datasets, evaluate model performance, and communicate results through visual reports that made the findings accessible to people who didn't know what an F1 score was but definitely cared whether the model was making good predictions.
- Strengthened my practical understanding of statistics, feature engineering, classification and regression workflows, and end-to-end data science execution — the internship connected concepts I'd learned in class to how they actually behave on real, imperfect data, which is where most of the learning happens.

Projects

Counterfactual Financial Scenario Generation Using Conditional Diffusion Models PyTorch, Time Series, DDPM

- Built a macro-conditioned Denoising Diffusion Probabilistic Model (DDPM) to generate multivariate financial return scenarios for SPY, QQQ, and GLD — conditioned the generation on interest rates, CPI, unemployment, Treasury yields, and VIX so the synthetic scenarios reflected realistic macroeconomic environments rather than just random noise that happened to look like historical returns.

- Designed the architecture with strict time-based alignment so training data didn't leak future information into the past, used rolling-window generation to produce coherent multi-step scenarios, and built temporal convolutional layers with timestep embeddings and macro-conditioning so the model understood both when and under what economic conditions each scenario was generated.
- Evaluated the generated scenarios on multiple dimensions — volatility error, cross-asset correlation error, tail-risk behavior, distributional realism, and autocorrelation diagnostics — because a model that gets the average right but misses the tails is dangerous in a financial context, and a model that generates scenarios where assets move independently is missing the whole point of portfolio analysis.

Real-Time Big Data Streaming Pipeline with Kafka and Spark Structured Streaming Kafka, Spark, PySpark

- Built a real-time streaming analytics pipeline using Kafka for ingestion and Spark Structured Streaming for processing — designed it for sensor-style event streams where data arrives continuously and you need answers in seconds, not hours. The pipeline handled producer-consumer workflows, schema parsing, and event-time-based processing so late-arriving data didn't silently corrupt the windowed aggregations.
- Implemented windowed aggregation, micro-batch execution, and output validation that verified the pipeline was producing correct results — streaming systems are harder to test than batch systems because the data never stops, so I built in checks that compared streaming outputs against periodic batch recomputations to catch drift early.

Vector Database Chatbot Using Chroma, Embeddings, and Python Chroma, NLP, Generative AI

- Built a retrieval-based chatbot using Chroma vector database and embeddings — the system chunked documents, stored them as vectors, and used semantic search to retrieve relevant context before generating responses. This architecture (RAG, retrieval-augmented generation) meant the chatbot could answer questions grounded in actual documents rather than hallucinating from model memory.
- Implemented the full pipeline — text ingestion, chunking strategies to keep related content together, vector storage and indexing, retrieval logic that balanced relevance with diversity, and response generation that incorporated the retrieved context. The trickiest part was tuning the chunk size: too small and the context was fragmented, too large and the retrieval lost precision.

Scalable Machine Learning Pipeline with PySpark PySpark, Spark SQL, ML Pipelines

- Built a PySpark-based machine learning workflow for data that was too large to fit in memory on a single machine — handled the full pipeline from exploratory analysis and feature preparation through model training and evaluation, all using Spark's distributed computing engine. The experience taught me that distributed ML adds complexity (partitioning, shuffling, serialization) but is essential when your dataset outgrows a single laptop.

Graph Analytics and Recommendation System Using Neo4j Graph Data Science Neo4j, Cypher, Recommendation Systems

- Modeled users, artists, and listening relationships as a graph in Neo4j — nodes for users and artists, edges for listening history and similarity — then applied graph analytics to extract patterns that a traditional relational approach would have needed multiple expensive joins to uncover. Built recommendation-oriented queries in Cypher that surfaced artists a user might like based on the listening behavior of similar users, which is essentially collaborative filtering expressed as graph traversal.

Additional Project Portfolio

Cloud-Based NLP Model for Automated Document Summarization — Built a summarization pipeline that could ingest documents and produce concise summaries using cloud-deployed NLP models.
Natural Language Processing Pipeline for Text Classification — End-to-end text classification workflow covering preprocessing, vectorization, model training, and evaluation across multiple model types.
Statistical Data Analysis and Visualization Using R — Applied statistical methods and created publication-quality visualizations in R for exploratory and confirmatory data analysis.
Parallel Data Processing and Machine Learning Workflow with Dask — Scaled ML workflows beyond single-machine memory limits using Dask and Dask-ML for parallel computation.
MNIST Classification with PCA, kNN, SVM, and Dimensionality Reduction — Classic MNIST digit classification comparing multiple techniques including PCA for dimensionality reduction, kNN, and SVM.
Dataset-Based Text Summarizer Using Machine Learning — Built an extractive text summarizer that ranked sentences by importance and generated summaries from source documents.
B2B E-Invoice Generator with QR Code and IRN Integration — Developed a full-stack invoicing system that generated QR-coded, IRN-compliant invoices for B2B transactions.

Achievements & Leadership

Served as Secretary of the Computer Science Department at Easwari Engineering College from 2021 to 2022 — bridged communication between faculty and 200+ students, coordinated departmental events, and kept academic activities running smoothly. The role taught me that good leadership is mostly about listening, following up, and making sure nobody's questions fall through the cracks. Also served as a Member of the Association of Computer Engineers during the same period, contributing to technical workshops, peer collaboration, and student-led initiatives that brought the department together outside of class. Across my academic projects, internships, and graduate coursework, I've built a strong applied portfolio spanning machine learning, data engineering, NLP, cloud analytics, statistical analysis, and software development — not just for grades, but because I genuinely enjoy building things that work.